

F878 — Lane A, the V_{us} tier seam (blind, one turn): my independent frame-product construction says $|V_{us}|$ MOVES with f ($\rightarrow$ Identified) and my blind “stable/Derived” prediction is FALSIFIED — BUT that same construction has a PHYSICAL DEFECT (mixing $\rightarrow 0$ for hierarchical weights, contradicting the observed 0.22), so it is NOT the faithful object. I cannot clear the seam from my side; the decisive two- f test must run in Elie’s brute-validated machine, and I specify exactly what it must show (f -stability) to earn Derived. Calibrated both ways: I do NOT force Derived (my number moves) and I do NOT force Identified (my construction is defective).

Lyra, Sunday 2026-08-09. Lane A, the 1-turn reconciliation. Pre-registered blind (F878_laneA_blind.md), computed, and the number falsified my prediction — then I caught that my construction itself is unphysical. RUBRICS below.

The seam

Scorecard carries $V_{us} = 1/\sqrt{20}$ as BOTH Derived (Elie’s validated FK machine, algebra==brute) and Identified (my F877, where the mixing moved with f). The directive’s test: hold f at two values, read $|V_{us}|$ — stable $\rightarrow$ Derived (down-template, $20 = \text{rank}^2 \cdot n_C$ target-innocent); moves $\rightarrow$ Identified.

What I computed (grace’s “mixing = d-vs-f eigenvector difference” object)

Built the frame-product $CKM = U_{up}^\dagger U_{down}$ on the shared ν -ladder (grace_CKM_localization: “the mixing lives in the d-vs-f eigenvector difference”; U_s diagonalizes $M_s = S \text{diag}(w_s) S^\dagger$, shared states S , down weight $d(\nu)$ fixed, up weight f swept).

f steepness s (up $\sim 90\times$ steeper than down)	$ V_{us} $
$s = 10$	0.0193
$s = 90$ (physical)	0.0020
$s = 300$	0.0006
$s = 1000$	0.0002
two physical f -values ($s=90$ vs 300)	$0.0020 \rightarrow 0.0006 = 70\% \text{ change} \rightarrow \text{MOVES}$
down-template limit $(U_{down})_{12}$	0.178 (vs $1/\sqrt{20} = 0.224$)

My blind prediction (stable $\rightarrow$ Derived) is FALSIFIED: in my construction $|V_{us}|$ moves 70% between two physical f -values and $\rightarrow 0$ as f steepens. From my side that reads Identified — and confirms F877.

BUT my construction has a PHYSICAL DEFECT (the honest catch that stops me capitulating either way)

My frame-product mixing $\rightarrow 0$ for hierarchical weights. That is physically wrong: the real up sector IS steeply hierarchical ($m_u:m_c:m_t \approx 1:600:80000$) AND the real quarks mix at $V_{us} = 0.22$. A faithful object must give $O(0.2)$ mixing *despite* the steep hierarchy. Mine doesn’t — so my construction ($M = S \text{diag}(w) S^\dagger$) is not the validated object, and its “moves $\rightarrow$ Identified” verdict is suspect on the same grounds F877 was (crude construction). I therefore do NOT bank Identified on it.

The reconciliation crux (what actually settles it — must run in Elie’s machine)

The seam is a CONSTRUCTION question: - If the physical V_{us} is a pure geometric overlap of the shared positions (the off-diagonal Gram element), it is independent of BOTH weights $d, f \rightarrow f\text{-stable} \rightarrow \text{Derived}$ (down-template $1/\sqrt{20} = 1/\sqrt{(\text{rank}^2 \cdot n_C)} = 1/(2\sqrt{n_C})$, target-innocent). - If it is the $d\text{-vs-}f$ eigenvector-frame misalignment (grace’s framing, my construction), it is $f\text{-sensitive} \rightarrow \text{Identified}$ — but my realization of that object has the wrong hierarchical limit, so it can’t be trusted to tier it. - Elie’s machine is the brute-validated one (algebra==brute, $V_{us}=1/\sqrt{20}$). The decisive test is to run the directive’s two- f test *there*: hold f at two physical values in the validated machine and read $|V_{us}|$. Citing $1/\sqrt{20}$ at ONE f is not sufficient — the tier turns on whether it is stable across f . If Elie’s validated $|V_{us}|$ is stable $\rightarrow \text{Derived}$ (and my frame-construction is simply the wrong object, defect confirmed). If it moves $\rightarrow \text{Identified}$ (and my F877/F878 were right).

Honest verdict (my side)

I cannot clear the seam with my construction — it moves (falsifying my prediction) but is physically defective (vanishing mixing for hierarchical weights). The target-innocent FORM $1/\sqrt{20} = 1/\sqrt{(\text{rank}^2 \cdot n_C)}$ supports Derived *if* Elie’s validated machine shows $f\text{-stability}$; my independent object cannot confirm that because it fails the hierarchical-mixing sanity check. Conservative scorecard tier until Elie’s two- f test: hold V_{us} at Identified-pending (do not upgrade the Cabibbo to Derived on one- f evidence). The number that clears Lane C is Elie’s two- f $|V_{us}|$ in the validated machine — I’ve specified exactly what it must show.

RUBRICS Layer-2 done-bar (Lane A, honest)

- ☒ Pre-registered blind; ran the two- f test; my prediction (stable/Derived) FALSIFIED — owned.
- ☒ Caught my construction’s physical defect (mixing $\rightarrow 0$ for hierarchical weights vs observed 0.22) $\rightarrow$ did NOT bank Identified on a defective object either.
- ☒ Located the crux (geometric-overlap vs eigenvector-frame object) and specified the decisive test (two- f in Elie’s brute-validated machine).
- ☒ Calibrated both ways; conservative interim tier (Identified-pending, not Cabibbo-Derived on one- f); magnitude discipline intact; no J value.

Handoffs

- @Elie (Lane A — you hold the decisive number) — my independent frame-product object MOVES with f (70% over two physical $f\text{-values}$) and $\rightarrow 0$ for steep f , which is physically wrong (real quarks mix at 0.22 despite steep hierarchy) — so my object is defective and can’t tier V_{us} . You must run the directive’s two- f test in your brute-validated machine: hold f at two physical values, report $|V_{us}|$ at each. Stable $\rightarrow$ Derived ($1/\sqrt{20} = 1/\sqrt{(\text{rank}^2 \cdot n_C)}$, target-innocent, and my frame-object is simply the wrong construction). Moves $\rightarrow$ Identified. One- f evidence (just “ $V_{us}=1/\sqrt{20}$ ”) does NOT settle it. Please also say what object your machine computes (direct overlap $\langle \text{up}_i | \text{down}_j \rangle$ vs eigenvector-frame product) — that IS the seam.
- @Keeper (Lane C/D) — the V_{us} tier is NOT yet reconciled: my construction says moves (Identified) but is defective; Elie’s says $1/\sqrt{20}$ but needs the two- f stability shown. Conservative interim: keep V_{us} at Identified-pending in v0.2.1 — do NOT flip CONDITIONAL PASS $\rightarrow$ PASS on the Cabibbo-Derived claim until Elie’s two- f number is in. I over-reached in my blind prediction (Derived) and the number corrected me; I’m holding the honest conservative tier.
- @Cal — I falsified my own blind Derived prediction AND refused to bank Identified on a defective construction; the decisive number is Elie’s two- f test in the validated machine. Magnitude off; no J value; $20 = \text{rank}^2 \cdot n_C$ confirmed target-innocent (clean form, IF stable).
- @Casey — the one-turn reconciliation didn’t fully close, and I want to be straight about why rather than paper it. I predicted, blind, that the Cabibbo number would hold steady when I wiggle the up-quark weighting — meaning it’s set by the down-template alone (derived). My own computation said the opposite: it moved a lot. But then I caught that my way of computing it has a bug — it makes quarks stop mixing when the masses get very spread out, and real quarks mix plenty despite very spread-out masses. So my “it moves”

verdict is on a broken calculator, and I won't trust it either. The clean resolution is one number from Elie's validated machine (the one that matches the brute-force integral): hold the up-weight at two values and see if the Cabibbo number stays at $1/\sqrt{20}$ or drifts. If it stays, it's derived (and $20 = \text{rank}^2 \times n_C$ is a clean, target-innocent form); if it drifts, it's one tier down. I've told him exactly what to run. I'd rather hand you an honestly-open seam with a sharp closing test than a forced "derived." Nothing pushed.

Notes only; no toy/theorem claimed (Lane A reconciliation, unresolved-from-my-side). F878: blind two-f V_{us} test. My frame-product object ($U_{up} \dagger U_{down}$, grace's d-vs-f eigenvector object): $|V_{us}|$ MOVES 70% over two physical f-values, $\rightarrow 0$ for steep f $\rightarrow$ my blind "stable/Derived" FALSIFIED. BUT construction is DEFECTIVE (mixing $\rightarrow 0$ for hierarchical weights contradicts observed 0.22) $\rightarrow$ did NOT bank Identified either. Crux: geometric-overlap object (f-independent $\rightarrow$ Derived) vs eigenvector-frame object (f-dependent $\rightarrow$ Identified); Elie's brute-validated machine is decisive. DECISIVE TEST (Elie must run): two-f $|V_{us}|$ in the validated machine — stable $\rightarrow$ Derived ($1/\sqrt{20} = 1/\sqrt{(\text{rank}^2 \cdot n_C)}$ target-innocent), moves $\rightarrow$ Identified; one-f insufficient. Interim: V_{us} Identified-pending; do NOT flip to Cabibbo-Derived on one-f. Calibrated both ways. Magnitude off. Nothing pushed. — Lyra